

JAKUB KIERZNOWSKI

Junior Java Backend Developer | Software Developer Intern at IBM

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github.com/qualv13 | linkedin.com/in/jakub-kierznowski | qualv13.github.io | instaldesk.pl

Java and Spring Boot backend developer. At IBM I work on Data Lineage, optimising graph and tree traversal over enterprise metadata. After hours I run InstalDesk, a B2B SaaS I built alone, with a native integration to the Polish KSeF e-invoicing system. Final-year Computer Science and Intelligent Systems at AGH, GPA 4.74/5.00, graduating January 2027.

EXPERIENCE

Software Developer Intern

IBM Software Lab · Kraków · hybrid
Part-time, 11/16 FTE (110 h/month)
Feb 2026 – Present

Backend work in Java and Spring on Data Lineage. I optimise graph and tree traversal over large enterprise metadata graphs. Wrote the automated pytest suite for that pipeline with parametrised cases and session-scoped fixtures; coverage went up and the CI run got shorter. I own the Jenkins job scheduling for it. Daily Scrum, English-speaking cross-functional team.

Solo Developer, InstalDesk

Own product · instaldesk.pl
Apr 2026 – Present

Quoting and VAT-invoicing SaaS for Polish installation contractors. A web PWA and a Windows desktop client share one account. Live, currently v1.9.0.

The KSeF integration is native: API v2 against the FA(3) schema, KSeF-token auth on web, XAdES-BES certificate auth on desktop, filed straight to the Ministry of Finance without a third-party integrator. Python and FastAPI on the backend, PostgreSQL in the cloud, SQLite locally so the desktop client keeps working offline. Per-tenant isolation, KSeF tokens encrypted at rest. Stripe billing over BLIK, card and Przelewy24, a 14-day trial without a card, renewal flows, transactional email, PDF generation, XLSX and CSV import and export. I built all of it and I support it.

IT Team Lead, Web & Workflow Automation

Student Council · WRSS WEAlIB, AGH
Mar 2025 – Present

Council member since Mar 2024, leading its IT team since Mar 2025. I maintain the council website (Vue, TailwindCSS) and the VPS it runs on, and built internal automations on self-hosted n8n against Slack and Google APIs over OAuth 2.0. Event coordinator for the VOLT 2025 tech conference, eight partner companies including Aptiv, PwC Polska and Cytiva, and part of the organising team for VOLT 2026.

Team Mentor and Community Leader

Zwolnieni z Teorii · remote
Sep 2022 – May 2025

Mentored 35+ student teams per edition, from planning to the final presentation of their social-impact projects. Later ran a participant community (ZWzT × PwC certificate) and supported olympiad entrants.

Game Developer

AGH Code Industry · Dec 2023 – Jun 2024

Gameplay mechanics in Unity and C# for “Student Trainer”, a game built by a student team under a mentor.

PROJECTS

DrugPath, biomedical knowledge-graph agent

neo4j-agent · Neo4j Aura Agent Hackathon 2026, top 6 · Neo4j Aura, Cypher, Python, LLM agents

An agent that walks the Hetionet biomedical graph (47,031 nodes, 293k relationships) along drug → gene → pathway → disease paths to explain how a drug works and where it might be repurposed. Educational project on public research data. Demo at qualv13.github.io/neo4j-agent.

IoT fleet backend

IoTServerApp · Java 17, Spring Boot 3.3, PostgreSQL, RabbitMQ/MQTT, Protocol Buffers, Docker

Backend for managing smart-lamp fleets. JWT auth with refresh tokens, MQTT command and telemetry topics over RabbitMQ, a Protocol Buffers device protocol, OTA firmware delivery from S3, scheduled device jobs. Layered Spring architecture with 15 JUnit 5 test classes covering controllers, services, repositories and security, integration tests on Testcontainers, and a full Docker Compose stack.

Green-window EV charging optimiser

CarCharging · Java, Spring Boot, REST, Docker

Finds the cleanest 48-hour charging window from the UK Carbon Intensity API with a sliding window over half-hourly forecasts, and serves the forecast through a REST API.

AI-native hardware rental and inventory hub

hardware-hub · Booksy technical assessment · Python, FastAPI, Vue, Gemini, Docker

Internal tool for tracking equipment, availability and hand-offs: semantic search over the inventory, an auditor that flags inconsistent records, and Gemini tool-calling for repairs. Every AI call has a deterministic fallback, so the product still works when the model does not.

BPM-first workout playlist generator

Fit2Beat · HackYeah 2025, top 8 of 65+ · Python, Flask, Docker

Builds playlists from a target BPM rather than from genre. Nobody else in the biohacking track approached it that way. Team of four, 24 hours, 3.9/5 from the jury.

Deep learning coursework

GGSN · PyTorch, TensorFlow, scikit-learn

Transfer learning with Grad-CAM explanations, DistilBERT sentiment classification, graph neural networks (GCN, GAT, GraphSAGE, GIN) over the Yelp dataset, and network compression by pruning, quantization and distillation across seven architectures.

SKILLS

Tech

Core. Java 17, Spring Boot, Spring Security, Hibernate/JPA, Maven, JUnit 5, Testcontainers, PostgreSQL, SQL, REST APIs.

Working knowledge. Python (FastAPI, pytest), Docker and Compose, Linux/Bash, Git, Jenkins CI/CD, RabbitMQ, MQTT, Protocol Buffers, OAuth 2.0, Neo4j and Cypher, AWS S3, algorithms and data structures.

Also used. Vue with TailwindCSS, C# (Unity), C++, n8n, LLM APIs, self-hosted models (Ollama), TensorFlow, scikit-learn.

Soft

Agile and Kanban, daily Scrum in a cross-functional English-speaking team. Team mentor at Zwolnieni z Teorii for three editions, 35+ teams each. Lead event coordinator for the VOLT 2025 conference, including partnership deals with tech companies. I ship my own product and handle its support and user contact myself.

Languages: Polish native, English C1 (working language at IBM), French A2.

EDUCATION

Computer Science and Intelligent Systems, BEng

AGH University · Kraków
Oct 2023 – Jan 2027

GPA 4.74/5.00. Year Representative since the 1st semester, Student Council since the 2nd. Engineering thesis: applying deep learning to 2D-image to 3D-mesh conversion.

Extended Mathematics, Physics and Computer Science

V High School · Kraków
2019 – 2023

C++, competitive-programming problem solving.

CERTIFICATIONS

Neo4j Certified Professional and Building Agents in Neo4j Aura, both 2026.

Amazon Junior Software Developer (Coursera specialisation): Data Structures and Algorithms, Programming with Java, Application Development.

INTERESTS

Snowboarding, sailing, a self-hosted homelab with Docker and quantised local models on a Raspberry Pi and a VPS, game dev, board games.

I hereby give consent for my personal data included in my application to be processed for the purposes of the recruitment process under Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC.